

Derivation of the Wallis' Integral

1 Aim

There had always been a pattern existing for the following integral

$$\int_0^{\frac{\pi}{2}} \sin^{10} x \, dx = \frac{9}{10} \cdot \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \quad (1)$$

It is also known as the "countdown formula". In this document we will verify the formula and provide the generalisation.

2 Proof

Let

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx \quad n \in \mathbb{N} \mid n \geq 2 \quad (2)$$

Using integration by parts

$$\text{Let } u = \sin^{n-1} x \quad (3a)$$

$$\text{Let } dv = \sin x \, dx \quad (3c)$$

$$du = (n-1) \sin^{n-2} \cos x \, dx \quad (3b)$$

$$v = -\cos x \quad (3d)$$

Thus

$$\begin{aligned} I_n &= \left[-\cos x \cdot \sin^{n-1} x \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \cos^2 x \cdot \sin^{n-2} x \, dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \cos^2 x \cdot \sin^{n-2} x \, dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \cdot \sin^{n-2} x \, dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \, dx - \int_0^{\frac{\pi}{2}} \sin^n x \, dx \\ &= (n-1) \left(\int_0^{\frac{\pi}{2}} \sin^{n-2} x \, dx - \int_0^{\frac{\pi}{2}} \sin^n x \, dx \right) \\ &= (n-1)(I_{n-2} - I_n) \\ nI_n &= (n-1) I_{n-2} \\ I_n &= \frac{n-1}{n} I_{n-2} \end{aligned} \quad (4)$$

Thus, for any $n \in \mathbb{N} \mid n \geq 2$, the integral of the n th power of sin can be summarised by equation 4.